Example 6.45

Verify whether the line $\frac{x-3}{-4}=\frac{y-4}{-7}=\frac{z+3}{12}$ lies in the plane $5x-y+z=8$.

Solution

Here, $(x_1,y_1,z_1)=(3,4,-3)$ and direction ratios of the given straight line are $(a,b,c)=(-4,-7,12)$. Direction ratios of the normal to the given plane are $(A,B,C)=(5,-1,1)$.

We observe that, the given point $(x_1,y_1,z_1)=(3,4,-3)$ satisfies the given plane $5x-y+z=8$

Next, $aA+bB+cC=(-4)(5)+(-7)(-1)+(12)(1)=-1\neq 0$. So, the normal to the plane is not perpendicular to the line. Hence, the given line does not lie in the plane. ■

6.8.8 Condition for coplanarity of two lines

(a) Condition in vector form

The two given non-parallel lines $\vec{r}=\vec{a}+s\vec{b}$ and $\vec{r}=\vec{c}+t\vec{d}$ are coplanar. So they lie in a single plane. Let A and C be the points whose position vectors are $\vec{a}$ and $\vec{c}$. Then A and C lie on the plane. Since $\vec{b}$ and $\vec{d}$ are parallel to the plane, $\vec{b}\times\vec{d}$ is perpendicular to the plane. So $\overrightarrow{AC}$ is perpendicular to $\vec{b}\times\vec{d}$. That is,

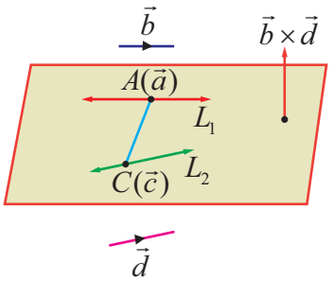


Fig. 6.30

$$(\vec{c}-\vec{a})\cdot(\vec{b}\times\vec{d}) = 0$$

This is the required condition for coplanarity of two lines in vector form.

(b) Condition in Cartesian form

Two lines $\frac{x-x_1}{b_1}=\frac{y-y_1}{b_2}=\frac{z-z_1}{b_3}$ and $\frac{x-x_2}{d_1}=\frac{y-y_2}{d_2}=\frac{z-z_2}{d_3}$ are coplanar if

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}=0$$

This is the required condition for coplanarity of two lines in Cartesian form.

6.8.9 Equation of plane containing two non-parallel coplanar lines

(a) Parametric form of vector equation

Let $\vec{r}=\vec{a}+s\vec{b}$ and $\vec{r}=\vec{c}+t\vec{d}$ be two non-parallel coplanar lines. Then $\vec{b}\times\vec{d}\neq\vec{0}$. Let P be any point on the plane and let $\vec{r_0}$ be its position vector. Then, the vectors $\vec{r_0}-\vec{a},\vec{b},\vec{d}$ as well as $\vec{r_0}-\vec{c},\vec{b},\vec{d}$ are also coplanar. So, we get $\vec{r_0}-\vec{a}=t\vec{b}+s\vec{d}$ or $\vec{r_0}-\vec{c}=t\vec{b}+s\vec{d}$. Hence, the vector equation in parametric form is $\vec{r}=\vec{a}+t\vec{b}+s\vec{d}$ or $\vec{r}=\vec{c}+t\vec{b}+s\vec{d}$.

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